Academic English Practice App - Agile Development Plan

1. Product Goal & Vision

Product Goal Statement:

> Build a mobile app that helps incoming university students transition from everyday English to academic English, enabling them to confidently participate in lectures, seminars, and academic discussions before starting their degree.

Target Users:

User Type	Characteristics	Key Needs
Incoming University Students	Non-native English speakers, Pre-university level	Build confidence in academic settings
	Anxious about classroom participation	Practice listening to lectures, academic vocabulary
	Limited time (prefer 5-15 min sessions)	Bite-sized, self-paced learning

Client Brief Alignment:

- ✅ Support key academic skills practice (listening, speaking, vocabulary)
- ✅ Improve confidence in lecture comprehension and seminar participation
- ✅ Enable progress tracking and self-paced learning
- ✅ GDPR compliant, secure storage of progress data

2. Team Structure & Role Rotation

Sprint	Week	Product Owner	Scrum Master	Development Team
1	3	Yanbin Xu	Yanshuo Liu	Junfan Zhou, Yupei Yang, Ruoqi Hu, Weike Jin, Chenxi Huang
2	4	Junfan Zhou	Yupei Yang	Yanbin Xu, Yanshuo Liu, Ruoqi Hu, Weike Jin, Chenxi Huang
3	5	Ruoqi Hu	Weike Jin	Yanbin Xu, Yanshuo Liu, Junfan Zhou, Yupei Yang, Chenxi Huang
4	6	Chenxi Huang	Yanbin Xu	Yanshuo Liu, Junfan Zhou, Yupei Yang, Ruoqi Hu, Weike Jin

Role Responsibilities:

Role	Key Responsibilities
Product Owner	Maintain backlog priority, accept completed work, represent client needs
Scrum Master	Facilitate ceremonies, remove blockers, track progress, update burndown
Developer	Implement features, write tests, participate in code review

3. Product Backlog (Ordered by Value)

Rank	ID	User Story	Priority	Points	Acceptance Criteria
1	US 01	As a learner, I want daily academic words with definitions so I can build vocabulary	P1	5	Display 5-8 words daily, English definitions, example sentences, audio pronunciation
2	US 02	As a learner, I want to save words to my bank so I can review later	P1	3	Add/remove words, review reminders, stored persistently
3	US 03	As a learner, I want to listen to lecture clips so I can practice comprehension	P1	8	Multiple clips, comprehension questions, difficulty levels marked
4	US 04	As a learner, I want AI pronunciation feedback so I can improve speaking	P1	8	Record voice, AI scoring, specific feedback on errors
5	US 12	As a learner, I want to track my progress so I can see improvement	P2	5	Dashboard showing completed exercises, scores, time spent
6	US 05	As a learner, I want vocabulary quizzes so I can test my retention	P2	5	Spelling, listening, fill-in-blank formats
7	US 06	As a learner, I want difficulty levels in listening so I can progress gradually	P2	5	Beginner/Intermediate/Advanced clearly labeled
8	US 07	As a learner, I want structured speaking practice so I can express ideas clearly	P2	8	Topic cards, 30-60s recording, AI feedback on fluency
9	US	As a learner, I want AI	P2	13	Office hour/group

Rank	ID	User Story	Priority	Points	Acceptance Criteria
	08	conversation practice so I can simulate real scenarios			discussion scenarios, voice I/O
10	US 10	As a learner, I want academic culture tips so I understand classroom norms	P3	5	Short videos, do's/don'ts cards, quiz
11	US 09	As a learner, I want free conversation mode so I can practice flexibly	P3	8	Open topic selection, natural conversation flow
12	US 11	As a learner, I want challenge mode so I can handle difficult situations	P3	8	Disagreement scenarios, follow-up questions

4. Sprint Plans

Week 2: Sprint 0 - Preparation

Activity	Tasks	Assignee	Output
Requirements Analysis	Refine user stories with acceptance criteria	All Team	Updated Backlog
Architecture Design	System design, database schema, API structure	Yanbin Xu, Yanshuo Liu	Architecture Doc
Technology Selection	Finalize tech stack (React Native, Node.js, OpenAI API)	Junfan Zhou, Yupei Yang	Tech Stack Doc
UI/UX Design	Wireframes for all core screens	Ruoqi Hu, Weike Jin	Figma Prototypes
Initial Planning	Estimate stories, plan Sprint 1	Chenxi Huang	Sprint 1 Plan

Sprint 1 (Week 3): Vocabulary Module

Sprint Goal: Deliver functional Vocabulary Hub where users can receive daily words and save to personal word bank

PO: Yanbin Xu | **SM**: Yanshuo Liu

Sprint Backlog

Task ID	Task	Priority	Assignee	Hours	Dependencies
T1.1	Database schema design & implementation	P1	Junfan Zhou	8	-
T1.2	Backend REST API framework setup	P1	Yupei Yang	10	T1.1
T1.3	User authentication (login/register)	P1	Ruoqi Hu	8	T1.2
T1.4	Daily Words feature (display, TTS audio)	P1	Weike Jin	10	T1.2
T1.5	Word Bank feature (add, delete, list, review)	P1	Chenxi Huang	8	T1.2, T1.3
T1.6	Main navigation UI components	P1	Yanbin Xu	8	-
T1.7	CI/CD pipeline setup	P2	Yanshuo Liu	6	-
T1.8	Unit tests for vocabulary APIs	P2	Junfan Zhou	4	T1.4, T1.5
T1.9	Integration testing	P2	Yupei Yang	4	T1.8

Realistic Capacity Check

- Total available hours: 7 members × 10 hrs/week = 70 hrs
- Planned work: 66 hrs ✅ (94% utilization, realistic)

Sprint 2 (Week 4): Listening + Speaking Basics

Sprint Goal: Deliver Listening Lab (Beginner/Intermediate levels) and Speaking Studio Level 1 (pronunciation practice)

PO: Junfan Zhou | **SM**: Yupei Yang

Sprint Backlog

Task ID	Task	Priority	Assignee	Hours	Dependencies
T2.1	Listening content database structure	P1	Yanbin Xu	6	-
T2.2	Audio player with playback controls	P1	Yanshuo Liu	8	-

Task ID	Task	Priority	Assignee	Hours	Dependencies
T2.3	Listening comprehension exercise UI	P1	Ruoqi Hu	8	T2.2
T2.4	AI-generated quiz question system	P1	Weike Jin	10	T2.1
T2.5	Word Quiz feature (spelling, listening)	P1	Chenxi Huang	6	Sprint 1
T2.6	Pronunciation recording & playback	P1	Junfan Zhou	8	-
T2.7	Speech recognition API integration	P1	Yupei Yang	10	T2.6
T2.8	Sentence follow-along practice	P2	Yanbin Xu	6	T2.7
T2.9	Unit & integration testing	P2	Yanshuo Liu	6	All above
T2.10	Sprint 1 bug fixes (from feedback)	P1	Ruoqi Hu	4	-

Sprint 3 (Week 5): Advanced Speaking + AI Conversation

Sprint Goal: Deliver Speaking Studio Level 2 (structured expression) and AI Conversation guided scenarios

PO: Ruoqi Hu | **SM**: Weike Jin

Sprint Backlog

Task ID	Task	Priority	Assignee	Hours	Dependencies
T3.1	Structured speaking with topic cards	P1	Yanbin Xu	8	-
T3.2	AI feedback system (fluency, accuracy, logic)	P1	Yanshuo Liu	10	T3.1
T3.3	Academic expression template library	P2	Junfan Zhou	6	-
T3.4	AI conversation dialog flow engine	P1	Yupei Yang	12	-

Task ID	Task	Priority	Assignee	Hours	Dependencies
T3.5	Guided scenario implementation	P1	Chenxi Huang	10	T3.4
T3.6	Voice input/output for AI chat	P1	Ruoqi Hu	8	T3.4
T3.7	Advanced listening (detail capture)	P2	Weike Jin	6	-
T3.8	Post-conversation report generation	P2	Yanbin Xu	6	T3.5
T3.9	System integration testing	P1	Yanshuo Liu	6	All above

Sprint 4 (Week 6): Integration & Polish

Sprint Goal: Complete all features, full system integration, deliver production-ready product

PO: Chenxi Huang | **SM**: Yanbin Xu

Sprint Backlog

Task ID	Task	Priority	Assignee	Hours	Dependencies
T4.1	Free Talk AI conversation mode	P2	Yanshuo Liu	8	Sprint 3
T4.2	Challenge Mode (difficult scenarios)	P3	Junfan Zhou	6	Sprint 3
T4.3	Academic Culture module (videos, tips)	P2	Yupei Yang	8	-
T4.4	Progress tracking dashboard	P2	Ruoqi Hu	6	-
T4.5	Full system integration testing	P1	Weike Jin	8	All modules
T4.6	User acceptance testing	P1	Chenxi Huang	6	T4.5
T4.7	Performance & security optimization	P1	Yanbin Xu	6	-
T4.8	Bug fixes and refinements	P1	All Team	8	Feedback

Task ID	Task	Priority	Assignee	Hours	Dependencies
T4.9	Documentation (user guide, API)	P2	Junfan Zhou	4	-
T4.10	Final demo preparation	P1	All Team	6	-

5. Agile Ceremonies

Daily Standup Format (15 min)

Each member answers:

1. What did I complete since last standup?
2. What will I work on next?
3. What blockers do I have?

SM updates burndown chart immediately after standup

Sprint Planning (2 hours, Day 1)

1. PO presents sprint goal
2. Team selects items from backlog
3. Tasks broken down and estimated
4. Capacity check performed
5. Commitment made

Sprint Review (1 hour, Last Day)

1. Demo working software to stakeholders
2. Collect feedback (documented in feedback log)
3. PO accepts/rejects completed items

Sprint Retrospective (1 hour, Last Day)

Question	Purpose
What went well?	Celebrate successes
What didn't go well?	Identify problems
What will we change?	Concrete improvement actions

Improvement actions tracked and reviewed next sprint

6. Definition of Done (DoD)

Criterion	Standard
Code Complete	All code written, follows conventions, merged to main
Code Reviewed	Reviewed by at least 1 other team member
Unit Tested	≥80% coverage, all tests pass
Integration Tested	Works correctly with other modules
UI/UX Verified	Matches design, responsive, accessible
Documentation	Code comments, API docs updated
No Critical Bugs	Zero P1 bugs, P2 bugs documented
PO Accepted	Meets acceptance criteria, PO signs off

7. Testing Strategy

Test Type	When	Responsible	Tools
Unit Testing	During development	Each developer	Jest, pytest
Integration Testing	End of each task	Assigned tester	Postman, Cypress
UI Testing	After feature complete	Ruoqi Hu, Weike Jin	Manual + Automated
User Acceptance Testing	Sprint 4	Chenxi Huang	Real user scenarios
Performance Testing	Sprint 4	Yanbin Xu	Load testing tools

8. Progress Tracking & Transparency

Visible Artifacts

Artifact	Location	Updated By	Frequency
Product Backlog	Shared board (Trello/Jira)	PO	As needed
Sprint Backlog	Sprint board	SM	Daily
Burndown Chart	Dashboard	SM	After each standup
Task Status	Board columns (To Do/In Progress/Done)	Each developer	Real-time

9. Feedback Response Process

Feedback Source	How Captured	Response Time	Action
Lecturer comments	Meeting notes	Same sprint	Add to backlog, prioritize
Client feedback	Demo feedback form	Next sprint	Refine requirements, adjust priorities
Team retrospective	Retro notes	Next sprint	Process improvements
Bug reports	Issue tracker	24 hours	Triage and assign

Evidence of change: Each sprint shows visible improvements from previous feedback

10. Risk Management

Risk	Probability	Impact	Mitigation	Owner
AI API integration delays	Medium	High	Early prototyping in Week 2, backup providers	Yupei Yang
Team member sick/unavailable	Low	Medium	Cross-training, pair programming, shared knowledge	SM
Scope creep	Medium	High	Strict backlog management, PO gatekeeping	PO
Underestimated complexity	Medium	Medium	20% buffer in estimates, daily risk review	SM
Technical skills gap	Low	Medium	Knowledge sharing sessions, online resources	All

Risks reviewed in every daily standup - raised early, never hidden

11. Client Acceptance Criteria

Based on client rubric, our product must meet:

Content Quality

Criterion	How We Ensure
English level suits learners	Content reviewed by team, progressive difficulty
Grammar/spelling accurate	Proofreading checklist, spell check
Examples support real English use	Academic scenarios from real university life

Criterion	How We Ensure
Instructions clear and simple	User testing with beginners

Learning Design

Criterion	How We Ensure
Learning path is logical	Week 1-2 → 3-4 → 5-6 progression built in
Teaches before tests	Tutorial content precedes assessment
Feedback after answers helpful	Specific feedback explaining right/wrong
Progress tracking useful	Dashboard showing growth over time

Usability

Criterion	How We Ensure
First use easy	Onboarding flow, minimal steps to start
Navigation easy	Consistent bottom nav, clear back buttons
Text readable	Minimum 16px font, high contrast
Buttons/labels clear	Icon + text labels, consistent styling
Layout consistent	Design system with reusable components
Error messages helpful	Friendly messages with next steps

12. Demonstration Checklist

Demo Structure

1. **Opening** (1 min): Clear product introduction
2. **Problem Statement** (1 min): Why this app is needed
3. **Target Users** (1 min): Who will use it
4. **Live Demo** (5-7 min): Realistic user scenario walkthrough
5. **Technical Choices** (2 min): Architecture and key decisions
6. **Agile Evidence** (2 min): Show backlog, progress, changes
7. **Testing Evidence** (1 min): How we verified quality
8. **Conclusion** (1 min): Value, limitations, next steps

Demo Scenario Example

> "Meet Sarah, a new international student starting next month. Watch her use the app to..."

Agile Evidence to Show

- Product backlog with priorities
- Sprint boards showing task movement
- Burndown charts from each sprint
- Retrospective improvement actions
- Feedback log and response evidence

13. Deliverables Summary

Sprint	Working Software	Documentation	Agile Artifacts
1	Vocabulary Module (Daily Words + Word Bank)	DB Schema, API Docs	Backlog, Sprint 1 Board, Burndown
2	+ Listening Lab + Speaking L1 + Word Quiz	Test Reports	Sprint 2 Board, Retro Notes
3	+ Speaking L2 + AI Guided Conversation	Integration Report	Sprint 3 Board, Feedback Log
4	Complete App (all features integrated)	User Guide, Final Report	All Boards, Demo Script

14. Team Member Responsibility Matrix

Member	Sprint 1	Sprint 2	Sprint 3	Sprint 4	Total Load
Yanbin Xu	PO, UI Design	Listening DB, Follow-along	Structured Speaking, Report	SM, Security	Balanced
Yanshuo Liu	SM, CI/CD	Audio Player, Testing	AI Feedback, Integration	Free Talk Mode	Balanced
Junfan Zhou	Database	Pronunciation	PO, Templates	Challenge Mode, Docs	Balanced
Yupei Yang	Backend API	SM, Speech API	Dialog Engine	Culture Module	Balanced
Ruoqi Hu	Auth System	Exercise UI, Bug Fix	PO, Voice I/O	Progress Dashboard	Balanced
Weike Jin	Daily Words	Quiz System	SM, Adv. Listening	Integration Test	Balanced
Chenxi Huang	Word Bank	Word Quiz	Guided Scenarios	PO, UAT	Balanced

Workload is distributed fairly - no single person carries more than 25% of any sprint